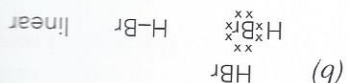
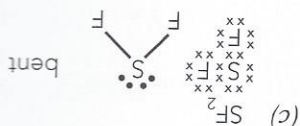
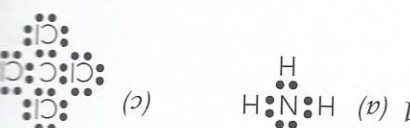
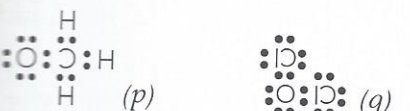
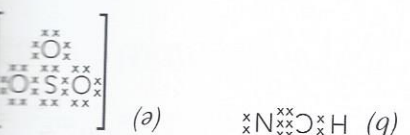
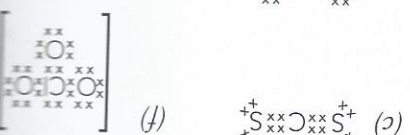
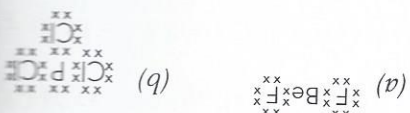


$$\% N = \frac{42.03}{149.10} \times 100 = 28.2\%$$



7.5 (a) done

7.4 F-F, C-S, B-H, N-O, N-H, C-O



Chapter Questions

$$= 0.515 \text{ kg}$$

$$= 2.43 \times \frac{21.2}{100}$$

(ii) mass of N in 2.43 kg of $(\text{NH}_4)_2\text{SO}_4$

$$= 2.43 \text{ kg}$$

$$= 2430 \text{ g}$$

$\therefore$ mass of fertiliser needed = (54) approximately.

(i)

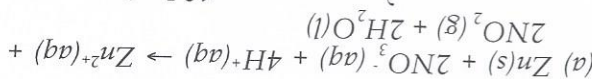
Urea, $\text{CO}(\text{NH}_2)_2$, has the highest weight.

$$\% N = \frac{60.06}{100} \times 100 = 46.7\%$$

(b) by proportion - to react all 12.2 g Na_2CO_3 we would need a total volume of:

$$(150 \text{ mL}) \times \frac{9.54}{12.2} = 192 \text{ mL of HNO}_3$$

$\therefore$ require a further 42 mL of HNO_3



$$125 \text{ mL}$$

$$6.0 \text{ mol L}^{-1}$$

$$n(\text{H}^+) = (6.0)(0.125) = 0.750 \text{ mol}$$

$$\therefore n(\text{Zn}) = (0.750)(1/4) = 0.1875 \text{ mol}$$

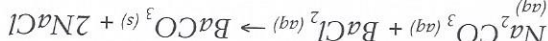
$$m(\text{Zn}) = (0.1875)(65.38) = 12.3 \text{ g}$$

$$(b) n(\text{Zn}(\text{NO}_3)_2) = (0.1875)(1/1) = 0.1875 \text{ mol}$$

$$m(\text{Zn}(\text{NO}_3)_2) = (0.1875)(189.4) = 35.5 \text{ g}$$

45.

(a) Yes, BaCO_3 will form.



$$200 \text{ mL}$$

$$1.0 \text{ mol L}^{-1}$$

$$200 \text{ mL}$$

$$1.0 \text{ mol L}^{-1}$$

$$(b) n(\text{Na}_2\text{CO}_3) = n(\text{BaCl}_2) = (1.0)(0.200) = 0.200 \text{ mol}$$

Hence no limiting reagent

$$\therefore n(\text{BaCO}_3) = (0.200)(1/1) = 0.200 \text{ mol}$$

$$\therefore m(\text{BaCO}_3) = (0.200)(197.3) = 39.5 \text{ g}$$

$$(c) c(\text{Ba}^{2+}) = c(\text{CO}_3^{2-}) = 0 \quad \text{All consumed}$$

$$c(\text{Na}^+) = \frac{\text{total moles}}{\text{total vol of solutions}}$$

$$= \frac{0.400}{(0.200)(2)}$$

$$= 1.0 \text{ mol L}^{-1}$$

$$\text{also } c(\text{Cl}^-) = \frac{0.400}{(0.200)(2)} = 1.0 \text{ mol L}^{-1}$$

For the Experts

46.

$$(a) M_r((\text{NH}_4)_2\text{SO}_4) = 132.15$$

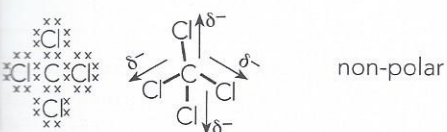
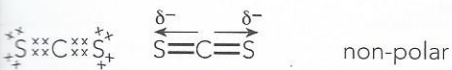
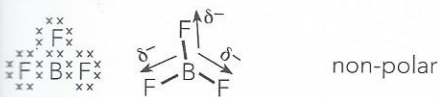
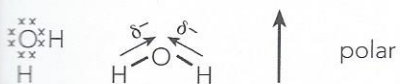
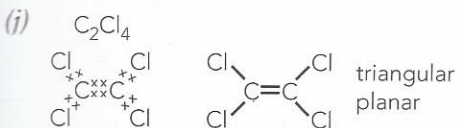
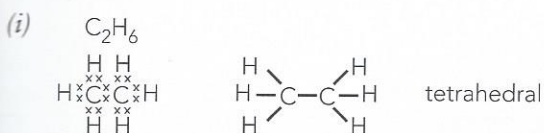
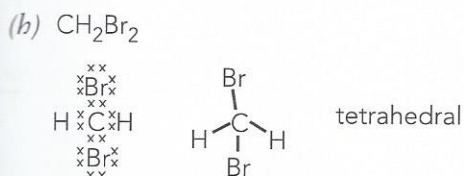
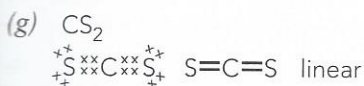
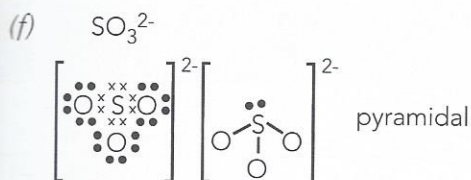
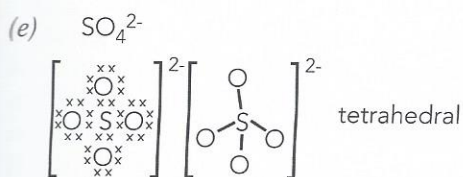
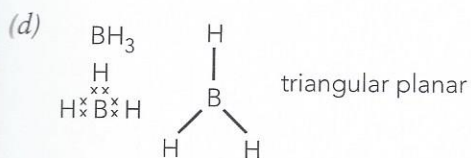
$$\% N = \frac{28.02}{132.15} \times 100 = 21.2\%$$

$$M_r(\text{NH}_4\text{NO}_3) = 80.5$$

$$\% N = \frac{28.02}{80.05} \times 100 = 35.0\%$$

$$M_r((\text{NH}_4)_3\text{PO}_4) = 149.10$$

$$\% N = \frac{42.03}{149.10} \times 100 = 28.2\%$$



7.7

(a) Pentane as it has dispersion forces only whereas 1-pentanol has hydrogen bonding. The weaker the intermolecular forces the greater the vapour pressure.

(b) Ethanol, water has stronger hydrogen bonding than ethanol.

(c) Pentane, both have dispersion forces but as the smaller molecule pentane will have weaker dispersion forces.

7.8

As non-volatile solute, the presence of Na^+ and Cl^- ions at the surface will reduce the number of water molecules that at the surface of the liquid and so reduce vapour pressure.

7.9

They would not be attracted to the molecules in the mobile phase and therefore not separate and move to different height up the stationary phase.

7.10

Molecules in component B have greater attraction for molecules in the stationary phase and do not move as far in the same time period.

7.11

Greater choice in the nature of the solid phase allows for greater range of mixtures to be separated and for greater degree of separation to occur in a shorter time. Paper not suitable for some corrosive materials.

7.12

(a) Paper or thin layer chromatography

(b) Gas chromatography

(c) High Performance Liquid Chromatography